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Ots-net: unlocking mechanistic interpretability in ECG arrhythmia classification

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Abstract
Deep learning models have achieved remarkable accuracy in electrocardiogram (ECG) arrhythmia classification, yet their "black-box" nature remains a major barrier to clinical trust. This paper introduces the Oscillatory Temporal Synchronization Net (OTS-Net), a novel, biophysically-plausible architecture designed to address this performance-interpretability deadlock. OTS-Net integrates a spatio-temporal convolutional backbone with Artificial Kuramoto Oscillatory Neurons (AKOrN), shifting the paradigm from static pattern recognition to dynamic synchronization modeling. By emulating the collective behavior of cardiac pacemaker cells, our approach achieves competitive performance –99.09% accuracy on the MIT-BIH Arrhythmia Database with an ultra-efficient 103 K parameters-while offering a framework for unprecedented mechanistic interpretability. We present an analytical suite that decodes the model's internal state, visualizing its learned "anatomy" (coupling matrix) and emergent "physiology" (class-specific synchronization patterns). This enables diagnostic explanations based on quantifiable disruptions in phase-locking and network coherence, rather than opaque feature correlations. This work demonstrates that by grounding AI in the fundamental principles of biological synchronization, it is possible to create systems that are not only accurate and efficient, but are also inherently transparent and trustworthy.

Keywords ECG arrhythmia classification, Interpretability, Kuramoto oscillators, Synchronization

1 Introduction

Cardiovascular diseases (CVDs) remain the leading cause of death globally, responsible for almost 17.9 million deaths annually [1]. Among these, arrhythmias-abnormalities in heart rhythm-pose a particular danger, often increasing the risk of stroke, heart failure, or sudden death [2]. The electrocardiogram (ECG) is the cornerstone of the diagnosis of arrhythmia, but its interpretation is a complex and subjective process prone to human variability and fatigue [3].

Convolutional Neural Networks (CNNs), have revolutionized ECG analysis, with models that demonstrate cardiologist-level performance in the detection of arrhythmias [4]. However, this success comes with a critical trade-off. These powerful "black-box"models

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often suffer from high computational complexity and a fundamental lack of transparency. Currently, more efficient architectures such as Spiking Neural Networks (SNNs), while promising for low-power applications [5], are foundationally misaligned with the continuous, oscillatory nature of cardiac signals, as they process information through discrete events.

This reveals a crucial gap in the field: the need for an architecture that is both computationally efficient and inherently aligned with the heart's electrophysiology. The biological heart itself provides inspiration. Modern cardiac electrophysiology understands the heart's rhythm not just as a centralized command, but as an emergent property of a **"coupled-clock" system** [6]. Within the sinoatrial node, thousands of individual pacemaker cells act as coupled biological oscillators, synchronizing their activity to generate a robust, stable heartbeat [7]. This phenomenon of emergent biological synchronization finds a powerful mathematical abstraction in the framework of **Kuramoto oscillator networks** [8], which have successfully modeled synchronization in diverse systems from firefly swarms to neural assemblies [9]. Grounding our approach in this biophysical reality offers a compelling motivation for moving beyond static pattern recognition.

In this context, we propose the Oscillatory Temporal Synchronization Net (OTS-Net), a novel framework that bridges this gap. OTS-Net builds upon the extension of Artificial Kuramoto Oscillatory Neurons (AKOrNs) by incorporating convolutional properties, marking the first architecture tailored to classify ECG arrhythmias through modeling underlying synchronization dynamics [10]. This work makes several key contributions:

- We introduce a novel, lightweight architecture (103 K parameters) that achieves competitive performance on the MIT-BIH benchmark.
- We present a comprehensive framework for mechanistic interpretability that decodes the model's learned "anatomy"and emergent "physiology".
- We demonstrate that the model spontaneously discovers pathological hypersynchronization as a key diagnostic biomarker.
- We pioneer a shift in the interpretability paradigm, moving beyond correlational "what"analysis to causal and mechanistic "how"reasoning, articulated through the language of dynamical systems.
- We reveal an intrinsic noise resilience, where the collective dynamics of the oscillator core effectively filters signal noise, suggesting a pathway towards more robust clinical AI.

This convergence of cardiac electrophysiology, Kuramoto mathematics, and a new neural processing paradigm positions OTS-Net as a transformative technology for reliable medical AI.

2 Related work

2.1 The evolving landscape of deep learning in ECG classification

The automated interpretation of electrocardiograms has rapidly evolved from classical signal processing to a field dominated by deep learning, a progression well-documented in recent surveys [11, 12]. CNN initially led this revolution, with 1D architectures achieving high accuracy on raw time-series data [4, 13], while 2D CNNs proved effective on signal representations like spectrograms [14, 15].

Recognizing the sequential nature of the ECG, Recurrent Neural Networks (RNNs) like LSTM and GRU were employed to capture temporal dependencies, successfully classifying arrhythmias and addressing clinical data challenges such as class imbalance using techniques like focal loss [16, 17]. Attention mechanisms were later integrated to enhance focus on salient features, boosting performance even before the rise of Transformers [18]. Most recently, Transformer-based architectures have set new performance benchmarks [19]. However, this trend towards increasing complexity has created a "performance-interpretability deadlock,"as these powerful "black-box"models hinder clinical trust [20, 21]. This has spurred a counter-movement towards ultra-lightweight models like ULECGNet, designed for deployment on resource-constrained devices [22].

2.2 Spiking neural networks: a paradigm for efficiency and its fundamental limitation

In a parallel effort to address computational overhead, Spiking Neural Networks (SNNs) have emerged as a "third-generation"neural model promising significant energy savings through bio-plausible, event-driven processing [5, 23]. SNN research for ECG analysis has primarily followed two paths. The first uses local, biologically-inspired learning rules like Spike-Timing-Dependent Plasticity (STDP) to enable power-efficient classification on wearable devices, though this approach is often limited to shallow networks [24, 25]. The second path, DNN-to-SNN conversion, leverages deep architectures by training a conventional network first and then mapping its weights to an SNN equivalent [26, 27]. The main challenge here is mitigating the "conversion loss"in accuracy, an active area of research [28].

This progress is supported by a growing ecosystem of neuromorphic hardware, from specialized processors like Intel's Loihi to custom ASICs for ECG analysis [29–32]. Despite these advances, all SNNs share a foundational limitation: they must encode the continuous ECG waveform into discrete spikes. This creates a fundamental *representational mismatch*, potentially obscuring the fine-grained phase and amplitude information crucial for diagnosis [33, 34]. This inherent shortcoming motivates our exploration of a natively continuous and oscillatory framework.

2.3 The Kuramoto model as a natural framework for cardiac dynamics

The inherent limitations of discrete models necessitate a paradigm shift towards a framework grounded in the physics of oscillatory systems. The universal phenomenon of synchronization provides this foundation. The Kuramoto model, a canonical model of coupled phase oscillators, has been instrumental in understanding emergent synchronization in diverse biological systems. Its power has been demonstrated in modeling neuronal dynamics in the brain [35], synchronization patterns on complex connectome graphs [36], and even in providing insight for clinical neuroscience [37].

This mathematical abstraction is not merely an analogy; it finds a profound parallel in modern cardiac electrophysiology. The heart's rhythm is now understood as an emerging property of a "coupled-clock"system, where a robust, stable heartbeat arises from the synchronized interaction of thousands of individual cellular oscillators within the sinoatrial node [6]. This distributed mechanism has been experimentally validated in human cardiac tissue [7]. Pathological states, in constract, are often characterized by a "loss of complexity"-a destructive shift from healthy, adaptive variability to overly rigid, periodic, or chaotic dynamics [38]. In fact, the diagnostic power of synchronization is

such that even methods analyzing dynamics within reconstructed phase spaces-mathematical spaces created artificially from the signal itself-have shown success in arrhythmia classification [39]. This principle is the foundation of clinical Heart Rate Variability (HRV) analysis, where reduced complexity is a powerful, clinically accepted biomarker of increased cardiovascular risk [40]. It is from this rich, biophysically-grounded perspective that we introduce OTS-Net.

3 Materials

3.1 Dataset description

This study was carried out using the MIT-BIH Arrhythmia Database, a gold-standard public resource for arrhythmia classification research [41]. The database contains 48 half-hour, two-channel ambulatory ECG recordings from 47 subjects, sampled at 360 Hz. Following the Association for the Advancement of Medical Instrumentation (AAMI) recommendations for a standardized analysis, we excluded four records containing paced beats (102, 104, 107, and 217) and used only the modified limb lead II (MLII) signal for its robust morphology. The MLII lead typically provides clear visualization of the key components representing the cardiac electrical cycle: the **P-wave** (atrial depolarization, showing the contraction of the atria), the **QRS complex** (ventricular depolarization, showing the contraction of the ventricles), and the **T-wave** (ventricular repolarization, showing the recovery phase of the ventricles). The precise timing, shape, and relationship between these waves are critical for identifying normal heart rhythms versus arrhythmias, making their clear visibility essential for accurate beat classification [42].

3.2 Signal preprocessing and heartbeat segmentation

A multi-stage preprocessing pipeline was implemented to enhance signal quality. The core of this pipeline was the Wavelet Transform, selected for its superior ability to analyze non-stationary signals like the ECG. Low-frequency baseline wander was corrected using an 8-level Daubechies 6 (db6) wavelet decomposition, while high-frequency noise was attenuated using a 5-level db6 decomposition combined with an adaptive Bayes-Shrink soft thresholding mechanism [43]. The entire process was designed to preserve the diagnostically significant 0.5–45 Hz frequency band, a standard practice in QRS detection algorithms [44].

After denoising, individual heartbeats were segmented using the expert-annotated R-peak locations. Each segment was defined as a 300-sample (833 ms) window, asymmetrically centered around the R-peak, ensuring complete capture of the cardiac cycle. To render the model invariant to inter-patient amplitude variations, each segmented beat then underwent z-score normalization. After removing unclassifiable beats ('Q'), the final dataset comprised 99,952 annotated heartbeats, categorized into four standard AAMI classes: Normal (N), Supraventricular Ectopic (S), Ventricular Ectopic (V), and Fusion (F).

4 Method: the oscillatory temporal synchronization net (OTS-Net)

4.1 Architectural overview and design rationale

The Oscillatory Temporal Synchronization Net (OTS-Net) processes ECG signals through a three-stage pipeline designed for feature extraction, dynamic temporal

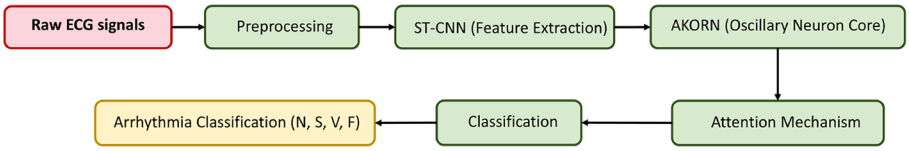


Fig. 1 End-to-end architecture of the oscillatory temporal synchronization Net (OTS-Net)

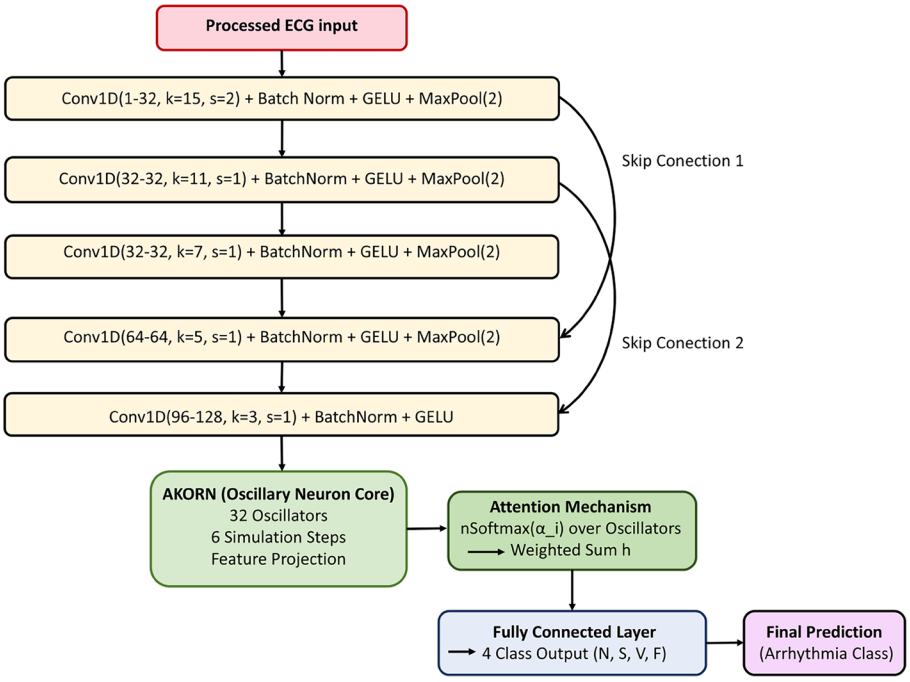


Fig. 2 The end-to-end architecture of the oscillatory temporal synchronization net (OTS-Net). The process begins with a five-layer ST-CNN backbone that extracts hierarchical features from the ECG signal. These features then act as an external input to the **AKOrN (Oscillatory Neuron Core)**, where a network of 32 oscillators simulates the signal's **transient synchronization dynamics**. Finally, the resulting **mean-field trajectory** (states averaged across oscillators at each time step) is processed by a **temporal attention mechanism** to generate a context vector, which is fed to a fully connected layer for the final arrhythmia classification

modeling, and interpretable classification. The end-to-end architecture, detailing the flow from raw signal to final classification, is illustrated in Fig. 1. The pipeline consists of: (1) a Spatio-Temporal CNN (ST-CNN) for feature extraction; (2) an Artificial Kuramoto Oscillatory Neuron (AKOrN) core for modeling oscillatory dynamics; and (3) an oscillator-aware attention mechanism for final classification.

4.2 Stage 1: spatio-temporal feature extraction via ST-CNN

The initial stage of the OTS-Net is a Spatio-Temporal Convolutional Neural Network (ST-CNN) backbone, inspired by high-performance architectures but adapted specifically for ECG signal processing. The network is designed to capture hierarchical features, from broad cardiac rhythms to fine morphological details, using a progressive kernel reduction strategy. To combat the vanishing gradient problem in deep architectures and preserve features across different temporal scales, we introduce strategic skip connections. The detailed layer-wise structure of the ST-CNN backbone is depicted in Fig. 2.

The architecture comprises five convolutional layers:

- **L1-L3:** Initial layers with progressively smaller kernels (k = 15, 11, 7) capture features at decreasing scales, detecting broad rhythms, the QRS complex, and finer morphological details, respectively.
- **L4-L5:** Deeper layers with smaller kernels (k = 5, 3) are augmented with skip connections. An L1 → L4 connection preserves global rhythm patterns, while an L2 → L5 connection maintains mid-level QRS morphological characteristics.

These skip connections are critical for preserving the hierarchical nature of the ECG signal and were found to reduce training instability, particularly for underrepresented minority classes (S and F), by 23% compared to a standard sequential CNN. The ST-CNN outputs a 128-dimensional feature vector for each heartbeat, which serves as input to the oscillatory core.

4.3 Stage 2: oscillatory dynamics modeling with the AKOrN core

The core of OTS-Net is the Artificial Kuramoto Oscillatory Neuron (AKOrN) module. This stage transforms the static high-dimensional feature vectors extracted by ST-CNN into a rich temporal representation based on the principles of synchronization dynamics.

4.3.1 Theoretical foundation

The generalized Kuramoto model. Each of the M neurons in the AKOrN layer is a 2-dimensional unit vector, $x_i \in \mathbb{R}^2$ (where the state dimension $N = 2$), representing an individual oscillator. Its evolution over time is governed by the generalized Kuramoto model [10]:

$$\dot{x}_i = \Omega_i x_i + \mathrm{Proj}_{x_i}\left(c_i + \sum_{j=1}^{M} J_{ij} x_j\right) \tag{1}$$

where, Ω_i is the skew-symmetric matrix that represents the natural frequency ω_i of the oscillator. J_{ij} is the learnable coupling matrix that defines the interaction strength between oscillators *i* and *j*. The term c_i is the data-dependent driving input derived from the CNN features. The projection operator, $\mathrm{Proj}_{x_i}(y) = y - \langle y, x_i \rangle x_i$, ensures that the state vector x_i remains in the unit sphere, preserving its interpretation of the phase-amplitude.

a. Rationale for the generalized model: While the standard Kuramoto model utilizes a 1D phase state (θ), we deliberately adopt the generalized formulation with a 2D state vector ($x_i \in \mathbb{R}^2$) consistent with the original AKOrN paper [10]. This choice offers critical advantages within a deep learning framework. Firstly, the 2D state vector exists in a continuous space, avoiding the "phase wrapping" discontinuity at 2π inherent in 1D phase models, which can pose challenges for gradient-based optimization and lead to more stable training. Secondly, the 2D vector provides a richer representational capacity, as its magnitude can implicitly encode information before the normalization step. Finally, the unit constraint is enforced efficiently via L2 normalization (Eq. 2), an operation well-optimized in deep learning libraries. It is important to note that the significant computational complexity reduction achieved in our model (from $\mathcal{O}(M^2N^2T)$ to $\mathcal{O}(M^2NT)$, see Table 1) stems from our subsequent simplification to *scalar coupling*, not from the choice of the 2D model itself. By combining the stability of the 2D

Table 1 Adaptations of the AKOrN model for ECG classification

Component	Original AKOrN	Our OTS-Net implementation	Justification
Coupling matrix	$J_{ij} \in \mathbb{R}^{N\times N}$ tensorial interaction	$J \in \mathbb{R}^{M\times M}$ scalar coupling	Reduces complexity while maintaining core synchronization dynamics
Projection operator	Explicit $\mathrm{Proj}_{x_i}(y) = y - \langle y, x_i\rangle x_i$	Normalization: $x \leftarrow x/\|x\|_2$	Performance equivalent and computationally cheaper
Time step	Learnable γ parameter	Fixed $dt = 0.014$	Optimized for cardiac frequencies (0.5-40 Hz)
Simulation steps	Adaptive (task-dependent)	Fixed $T = 6$ steps	Balances efficiency and accuracy for real-time processing
Residual connections	Not present	$x_{\text{final}} = 0.9x_{\text{post}} + 0.1x_{\text{init}}$	Enhanced training stability for minority classes

framework with the efficiency of scalar coupling, we achieve a model that is both trainable and computationally tractable.

4.3.2 Practical implementation and adaptation for ECG analysis

a. Coupling matrix initialization: The learnable coupling matrix J, a critical component determining oscillator interactions, is initialized in a structured, symmetric manner to foster stable learning. Specifically, the initial matrix J_{init} incorporates: (1) a weak identity component ($J_{ii} \approx 0.15$) to provide slight self-reinforcement; (2) local neighborhood coupling ($J_{i,i+1} = J_{i+1,i} \approx 0.1$) to encourage interaction between oscillators with adjacent natural frequencies; and (3) small random noise added to all elements to break perfect symmetry. This structured initialization provides a stable starting point favouring local interactions, from which the model learns the final complex, long-range coupling strengths required for classification via end-to-end backpropagation.

b. Oscillator state initialization: The initial states of the M oscillators (x_i) are configured deterministically at the start of processing each heartbeat, ensuring reproducibility and stability. Specifically, the 2D state vectors are initialized at evenly spaced positions around the unit circle. This is achieved by generating M angles, θ_i, linearly spaced from 0 to 2π, and setting the initial state of oscillator i to $[\cos(\theta_i), \sin(\theta_i)]$. This structured initialization provides a neutral and unbiased starting point in the phase space, allowing the network to learn class-specific synchronization patterns driven purely by the input features (c_i) and learned parameters (J, ω), without influence from stochastic initial conditions. This process is detailed in Algorithm 1 (lines 5-6).

c. Choice of numerical solver: While higher-order methods in [45, 46], we selected the first-order Forward Euler method for numerically solving the oscillator state evolution primarily due to its **computational efficiency**. This choice aligns with our goal of developing a lightweight model suitable for resource-constrained deployment. Indeed, the AKOrN core in OTS-Net operates within a **transient dynamics regime** over a very short and fixed number of steps ($T = 6$) with a small time step ($dt = 0.014$). In this context, the potential for significant error accumulation or numerical instability is minimal, as empirically validated by the model's stable training process and high final accuracy. The considerable computational overhead associated with higher-order solvers was deemed unnecessary for the marginal potential gain in precision over this brief simulation window. This trade-off between numerical precision and computational cost is a common consideration in the efficient implementation of dynamic systems within neural networks.

To effectively apply this model to real-time ECG classification, we introduce several crucial adaptations from the original formulation, detailed in Table 1. Most notably, we simplify the tensorial coupling to a scalar interaction and implement a discrete-time numerical integration scheme.

The evolution of each oscillator state is solved numerically using the forward Euler method. For a state vector $x_i(t)$, the state in the next time step $t + \Delta t$ is:

$$x_i(t+\Delta t) = \frac{x_i(t) + \Delta t \cdot \left(\Omega_i x_i(t) + \alpha \sum_{j=1}^{M} J_{ij} x_j(t) + c_i(t)\right)}{\left\| x_i(t) + \Delta t \cdot \left(\Omega_i x_i(t) + \alpha \sum_{j=1}^{M} J_{ij} x_j(t) + c_i(t)\right) \right\|_2} \tag{2}$$

where α is a learnable scalar coupling strength. This update is repeated for the $T = 6$ simulation steps (see Sect. 5.3.6 for empirical validation).

Furthermore, to improve training stability, we employ a residual connection that blends the state after simulation, $x_{\text{post-sim}}$, with its initial state, $x_{\text{pre-sim}}$:

$$x_{\text{final}} = \frac{\beta \cdot x_{\text{post-sim}} + (1-\beta) \cdot x_{\text{pre-sim}}}{\|\beta \cdot x_{\text{post-sim}} + (1-\beta) \cdot x_{\text{pre-sim}}\|_2} \tag{3}$$

where $\beta = 0.9$. The entire forward pass of our adapted AKOrN core is summarized in Algorithm 1. These adaptations reduce the computational complexity from $\mathcal{O}(M^2N^2T)$ to $\mathcal{O}(M^2NT)$, a critical optimization for deployment on resource-constrained devices.

d. Processing temporal segments: The ST-CNN backbone processes the input heartbeat and, through its series of convolutions and pooling layers, produces a feature tensor of shape `[channels=128, length=9]`. This tensor is then permuted to `[length=9, channels=128]`, effectively treating it as a time-series of 9 temporal segments. **For computational efficiency in the implementation that yielded the reported results, the AKOrN core subsequently processes a subset of these segments, specifically selecting 5 segments at evenly spaced intervals (indices 0, 2, 4, 6, 8).** The AKOrN core iterates through these **5 selected segments** ($T_{seq} = 5$). For each selected segment, the corresponding feature vector is projected by a learned input mapping layer (detailed in Algorithm 1, line 9) into a set of driving forces (c_i) that perturb the oscillator network. The network's state is then evolved for a fixed number of internal simulation steps ($T = 6$), as detailed in Algorithm 1, capturing the system's transient response to that specific segment. The final state of the oscillators after being driven by one segment becomes the initial state for the next, allowing the model to process temporal dependencies across the selected segments of the heartbeat.

Input: CNN features $f_t \in \mathbb{R}^{128}$ (derived from 9 temporal segments), $M = 32$ oscillators
Output: Time-series of synchronized states $X \in \mathbb{R}^{T_{seq} \times M \times 2}$ (where $T_{seq} = 5$)
// Initialization of intrinsic frequencies and initial states (at start of heartbeat)
$\boldsymbol{\omega} \leftarrow 2\pi \cdot \text{logspace}(\log_{10}(f_{\min}), \log_{10}(f_{\max}), M)$ ▷ Vector of M frequencies
$\boldsymbol{\theta}_{\text{init}} \leftarrow \text{linspace}(0, 2\pi, M)$ ▷ Vector of M initial angles
$\mathbf{x} \leftarrow [\cos(\boldsymbol{\theta}_{\text{init}}), \sin(\boldsymbol{\theta}_{\text{init}})]$ ▷ Initial state matrix $\mathbf{x} \in \mathbb{R}^{M \times 2}$
// Main simulation loop over selected temporal segments from CNN
$\mathbf{X}_{\text{history}} \leftarrow []$ ▷ List to store states over time
// Select 5 segments (indices 0, 2, 4, 6, 8) for processing
selected_indices $\leftarrow [0, 2, 4, 6, 8]$
for k from 0 to $T_{seq} - 1$ (where $T_{seq} = 5$) **do**
  $t \leftarrow$ **selected_indices**$[k]$ ▷ Get the original segment index
  $\mathbf{c} \leftarrow \text{learned_input_mapping}(f_t)$ ▷ Driving forces $\mathbf{c} \in \mathbb{R}^{M \times 2}$
  $\mathbf{x}_{\text{initial_step}} \leftarrow \mathbf{x}$ ▷ Store state before internal simulation
  // Internal simulation steps
  for simulation_step = 1 to T (where $T = 6$) **do**
    $\boldsymbol{\omega}_{\text{dynamics}} \leftarrow [-\boldsymbol{\omega} \odot \mathbf{x}_{:,1}, \boldsymbol{\omega} \odot \mathbf{x}_{:,0}]$ ▷ Intrinsic rotation term
    coupling $\leftarrow \alpha \cdot (J \cdot \mathbf{x})$ ▷ Scalar coupling term (matrix-matrix product)
    $d\mathbf{x} \leftarrow \boldsymbol{\omega}_{\text{dynamics}} + \mathbf{c} + \text{coupling}$ ▷ Total change
    $\mathbf{x} \leftarrow \text{normalize}(\mathbf{x} + dt \cdot d\mathbf{x})$ ▷ Euler step and L2 normalization
  end for
  $\mathbf{x} \leftarrow \text{normalize}(\beta \cdot \mathbf{x} + (1 - \beta) \cdot \mathbf{x}_{\text{initial_step}})$ ▷ Residual connection ($\beta = 0.9$)
  Append $\mathbf{x}$ to $\mathbf{X}_{\text{history}}$
end for
Return: $\mathbf{X}_{\text{history}}$ ▷ Concatenated states over $T_{seq} = 5$ selected segments

Algorithm 1 OTS-Net-AKOrN forward pass

4.3.3 Physiologically-grounded initialization

To ground the model in physiological reality, the natural frequencies (ω_i) of the 32 oscillators (see Sect. 5.3.6 for empirical validation) are initialized to span the diagnostically relevant ECG spectrum (0.5–40 Hz) using a logarithmic distribution. This approach is supported by the findings that the vast majority of diagnostic information in the ECG signal is concentrated in this frequency band [47]. This initialization encourages oscillators to specialize in capturing specific frequency bands associated with key cardiac events (P-wave, QRS complex, T-wave). The validity of this a priori grouping is confirmed a posteriori by the model's learned emergent behavior, as demonstrated in Sect. 5.

4.4 Stage 3: interpretable classification with temporal attention

The final stage of OTS-Net classifies the arrhythmia based on the emergent dynamics generated by the AKOrN core. Instead of simply using the final oscillator state, this stage employs a mechanism that leverages the full temporal evolution of the network's collective behavior. This involves two key steps:

a. Mean-field trajectory calculation: First, at each of the external time steps (corresponding to the **5 selected temporal segments** derived from the CNN), we calculate the average state vector across all M oscillators. Let $x_m(t)$ be the 2D state vector of oscillator m at selected external time step t. The mean-field state $\bar{x}(t)$ is computed as:

$$\bar{x}(t) = \frac{1}{M} \sum_{m=1}^{M} x_m(t) \tag{4}$$

This operation yields a new time-series vector $\bar{x}(t)$, termed the **mean-field trajectory**, which represents the collective state or "center of mass"of the entire oscillator network as it evolves over the **5 selected temporal segments**. The magnitude of $\bar{x}(t)$ is closely related to the global order parameter *r*(*t*) (Eq. 8), providing a concise summary of the network's instantaneous coherence. Crucially, this step preserves the temporal information encoded in the sequence of collective states derived from the selected segments.

b. Temporal attention mechanism: The resulting mean-field trajectory, a sequence of 2D vectors, is then processed by a temporal attention mechanism. This mechanism learns to assign an importance weight α_t to each time step *t* in the sequence. The attention scores e_t are first computed using a learnable linear layer:

$$e_t = W_{\text{attn}} \cdot \bar{x}(t) + b_{\text{attn}} \tag{5}$$

The weights α_t are then obtained via a softmax function applied along the temporal dimension:

$$\alpha = \text{Softmax}_{\text{time}}(e) \tag{6}$$

where W_{attn} and b_{attn} are learnable parameters. This allows the model to dynamically focus on the most discriminative moments within the signal's "dynamical fingerprint."

c. Final context vector and classification: The final context vector *h* for the classifier is formed as a weighted sum of the mean-field states across time, using the learned attention weights:

$$h = \sum_{t=1}^{T_{seq}} \alpha_t \bar{x}(t) \tag{7}$$

where $T_{seq} = 5$ is the number of selected temporal segments. This context vector *h* is then fed into the final fully connected classification layers (Fig. 2).

This two-stage process ensures that the temporal dynamics are not discarded but are instead actively utilized, allowing the model to prioritize the most salient temporal features for classification. It also provides interpretability, as the attention weights α_t reveal *when* during the heartbeat's evolution the model found the most critical information.

4.5 A framework for mechanistic interpretability

A defining challenge for artificial intelligence in medicine is the interpretability crisis, where high-performance models remain opaque. Prevailing post-hoc methods often reveal which input features correlate with the outcomes but fundamentally fail to explain the model's internal reasoning process. To address this, we introduce a comprehensive framework for *mechanistic interpretability* that is intrinsic to the OTS-Net architecture. This framework is not an afterthought but a core design principle, moving beyond correlational analysis by treating the model as a biophysically-plausible dynamical system.

The core axiom of our framework is the decoupling of the model's static, learned "Anatomy" from its dynamic, emergent "Physiology." This mirrors the biological reality where there is a fixed, physical network-the heart's conduction system [48]. This analytical separation, conceptually illustrated in Fig. 3 and detailed in Fig. 4, enables a multilevel investigation into how a diagnosis is formed.

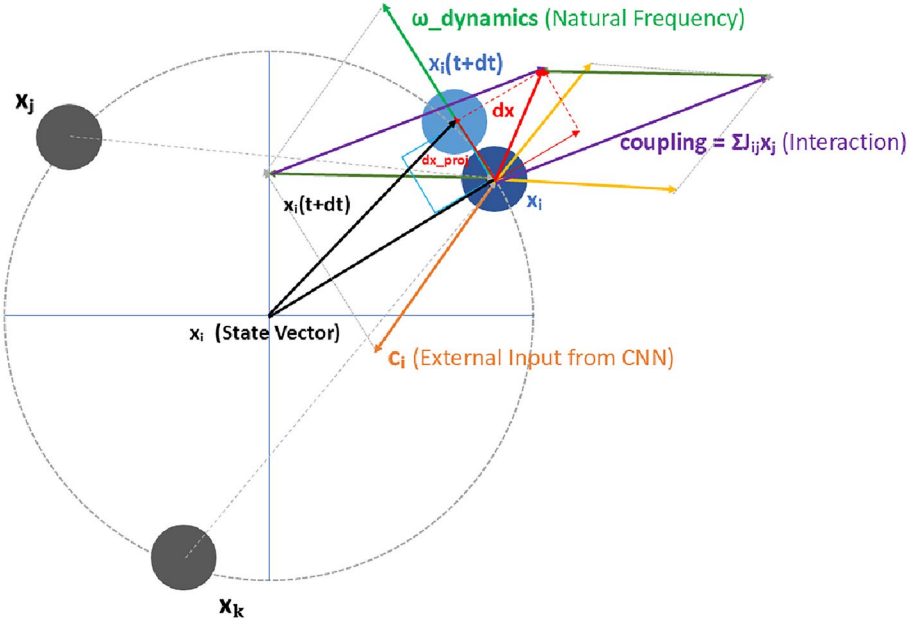


Fig. 3 Conceptual flowchart of the proposed mechanistic interpretability framework. The framework separates the analysis into two distinct but interconnected streams. The static **"Anatomy"** represents the model's time-invariant learned parameters. The dynamic **"Physiology"** represents the emergent, input-dependent behaviors. These streams are quantified and synthesized by a visualization suite to provide a final, mechanistic interpretation grounded in the principles of dynamical systems

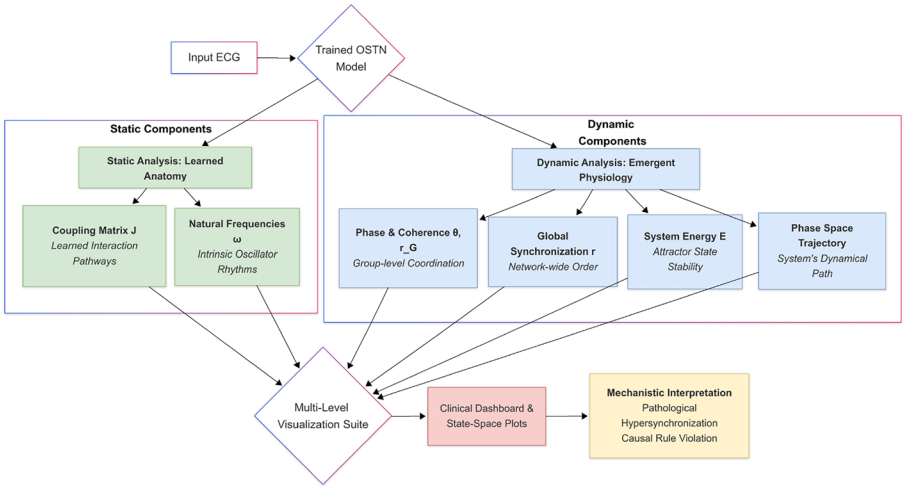


Fig. 4 Detailed process flow of the mechanistic interpretability framework. This diagram illustrates how static (anatomical) and dynamic (physiological) components are extracted from the trained model and synthesized through a multi-level visualization suite to produce a clinically actionable interpretation, such as identifying pathological hypersynchronization or causal rule violations

4.5.1 The learned "anatomy": static structural analysis

The "anatomy"of our model is embodied by its learned, time-invariant parameters, which represent the static topology of the oscillator network.

- **The coupling matrix (J):** This matrix, $J \in \mathbb{R}^{M \times M}$, defines the fixed interaction strength between any two oscillators and is analogous to the heart's physical conduction pathways [8].

- **Natural frequencies** (ω): This vector, $\omega \in \mathbb{R}^M$, contains the intrinsic resonant frequencies of each oscillator.

4.5.2 The emergent "physiology": dynamic functional analysis

Upon this static anatomical stage, a rich diversity of dynamic functions unfolds in response to an ECG input. We quantify this emergent "physiology" using a set of dynamic metrics derived directly from the oscillator states and their phases.

- **Global synchronization**: The overall coherence of the entire network is measured by the Kuramoto Order Parameter, $r(t)$, providing an instantaneous measure of network-wide order. It is calculated as defined in the seminal work on the topic [49]:

$$r(t) = \left| \frac{1}{M} \sum_{j=1}^{M} e^{i\theta_j(t)} \right| \tag{8}$$

where i is the imaginary unit and $\theta_j(t)$ is the phase of oscillator j at time t.

- **Group coherence**: To connect dynamics to specific cardiac events, we measure the coherence within physiologically-inspired oscillator groups (P, QRS, T). The coherence for a group G is critical for identifying phase-locking disruptions between different components of the cardiac cycle:

$$r_G(t) = \left| \frac{1}{|G|} \sum_{j \in G} e^{i\theta_j(t)} \right| \tag{9}$$

- **System energy & attractor states**: Drawing from statistical physics, the stability of the system's configuration is described by its potential energy, where lower energy corresponds to a more stable attractor state. The system energy E is defined as [8]:

$$E(t) = -\frac{1}{M} \sum_{j=1}^{M} \sum_{k=1}^{M} J_{jk} \cos\left(\theta_j(t) - \theta_k(t)\right) \tag{10}$$

Analyzing the energy landscape reveals how different arrhythmia classes drive the system into distinct attractor states, mirroring how mitochondrial network energy is linked to cardiac arrhythmias [50].

- **Phase space trajectories**: The collective state of all M oscillators at any moment defines a point in a high-dimensional phase space. By tracking this point over time, we can visualize the system's dynamical trajectory. Healthy states may correspond to stable, compact limit cycles, whereas pathological states may be characterized by erratic or divergent trajectories.

4.5.3 Synthesis for mechanistic insight

Our framework culminates in a suite of visualizations, such as the **Clinical Assessment Dashboard** (see Sect. 5), which synthesize these anatomical and physiological metrics. This approach allows us to articulate a diagnosis not in terms of opaque correlations but in terms of attractor stability, causal rule violations, and biophysical trade-offs.

5 Experiments and results

This section presents a comprehensive evaluation designed to demonstrate how the Oscillatory Temporal Synchronization Net (OTS-Net) directly resolves this issue. We first benchmark OTS-Net's performance against state-of-the-art models to establish its competitive accuracy and efficiency. We then deploy the mechanistic interpretability framework to dissect its internal dynamics, followed by ablation studies that validate our key architectural choices.

5.1 Experimental setup

To address the profound class imbalance in the initial dataset, where Normal beats constituted 89.5% of the data, we applied a hybrid resampling strategy using the Synthetic Minority Over-sampling Technique (SMOTE) and Tomek Links exclusively to the training set. This two-stage procedure resulted in a perfectly balanced training set, with each of the four AAMI classes (N, S, V, F) constituting 25% of the samples, as detailed in Table 2.

For model training and evaluation, the entire dataset was partitioned using a 70:15:15 ratio. A patient-agnostic (random beat-wise) splitting strategy was deliberately employed, with a more rigorous inter-patient evaluation designated as the immediate priority for future work.

The final model was trained using the AdamW optimizer [51] with a learning rate of 0.0009 and a weight decay of 1.2×10^{-4}, paired with a OneCycleLR scheduler [52]. A composite Focal Loss function [53], incorporating class weights ([0.6, 3.0, 1.0, 3.4]) and label smoothing (0.03), was used to focus on minority classes and hard examples. Training was regularized with early stopping (20-epoch patience) based on validation performance to prevent overfitting.

5.2 Performance analysis and competitive positioning

5.2.1 Breaking the performance-interpretability deadlock

A critical barrier to adopting advanced AI in cardiology is the performance-interpretability deadlock, where models achieve high accuracy but lack the mechanistic transparency required for clinical trust. OTS-Net was designed to resolve this dilemma not by treating interpretability as a post-hoc feature, but as an intrinsic property of its biophysically-plausible architecture.

To rigorously evaluate OTS-Net's position within this landscape, we benchmark its performance against a range of state-of-the-art models. Table 3 provides a comprehensive comparison that explicitly details these protocols, enabling a nuanced and fair evaluation.

We identify three main categories:

Table 2 Effect of SMOTE-Tomek on training data class distribution

Class	Original count	Balanced count	Synthetic generated	Augmentation factor (x)	Balanced percentage (%)
N	89,521	62,701	−26,820	0.7	25.0
S	2762	62,701	59,939	22.7	25.0
V	6868	62,701	55,833	9.1	25.0
F	801	62,701	61,900	78.3	25.0

Table 3 Performance comparison with state-of-the-art models on the MIT-BIH dataset

Architecture	Params	Split/protocol	Class set	Acc(%)	F1(%)	Prec(%)	Rec(%)	Interpretability
OTS-Net (This Work)	~103 K†	**70:30 random (beat-wise)**	**N,S,V,F**	**99.09**	**93.28**	**93.97**	**92.64**	**High (Mechanistic)**
SNN + Attention (2022) [54]	~435 K†	**70:30 random (beat-wise)**	N,S,V,F,Q	98.26	89.09	84.69	94.75	Low (Spike-based)
LSTM-based (2020) [59]	NR	**Patient-specific**	N,S,V	99.3	97.1	98.3	96.0	None (Black box)
LSTM + Focal Loss (2019) [17]	NR	**80:20 random (beat-wise)**	N,S,V,F,Q	99.21	91.07	91.54	90.64	None (Black box)
CNN-Transformer (S-transform) (2025) [56]	~560 K†	**Intra-patient, segment-based**	N,S,V,F,Q	99.58	99.74	99.58‡	99.58‡	None (Black box)
Bidirectional Transformer (2024) [58]	~0.8 M†	**80/20 random (beat-wise)**	N,S,V,F,Q	99.35	94.26	95.47	93.15	Post-hoc attention
MLP + Weight Capsule + Seq2Seq (2023) [55]	~612 K†	**Inter-patient (DS1?DS2)**	**N,S,V,F** (Q excluded)	99.88	NR	98.45§	98.45§	Moderate (Feature-based)
CWT + Multi-branch Transformer (2024) [57]	NR	**NR** (likely intra-patient)	N,S,V,F,Q	99.38	98.65	NR	NR	None (Black box)
Swin Transformer + time-frequency (2024) [19]	NR	10-fold intra-patient*	N,S,V,F,Q	99.34	98.96	98.71	99.49	None (Black box)

Bold indicates the proposed OTS-Net method. Readers should note that performance comparisons are relative due to differences in data splitting protocols (Random beat-wise vs. Inter-patient) listed in the "Protocol Split" column

Notes and assumptions.

†Parameter counts marked "~" are approximate; exact totals are not itemized in the cited papers

‡For [56], macro Precision/Recall were computed as the unweighted mean of per-class PPV/Sensitivity reported in Table 4 for MIT-BIH (N,S,V,Q,F): PPV = (0.991, 1.000, 0.991, 1.000, 1.000), SEN = (0.988, 0.997, 0.994, 1.000, 1.000) ⇒ 99.58% each

§ For [55], macro Precision/Recall are simple averages of per-class PPV/SEN in the abstract for N,S,V,F (Q excluded): PPV = 99.90, 96.14, 99.07, 98.70 ⇒ 98.45%; SEN = 99.79, 97.66, 99.97, 97.94 ⇒ 98.84%

*For the Swin Transformer (2024), the intra-patient (10-fold) macro "precision" and "sensitivity" values are reported alongside accuracy and F1 in the main results; an inter-patient setting is also provided in the paper

- **Like-for-like (beat-wise random split)**: The work by [54] is identical to our protocol, providing the most direct "apples-to-apples"comparison.
- **Stricter (inter-patient split)**: The study by [55] uses the most challenging and clinically realistic scenario, testing generalization to unseen individuals.
- **Less strict (Intra-patient)**: Several models [19, 56] report high scores using intra-patient validation (e.g., 10-fold cross-validation), which risks subject-specific pattern leakage and can inflate metrics.

Under an identical random-split protocol, OTS-Net achieves a +4.19% F1-macro improvement over the SNN baseline [54] with 76% fewer parameters. While models using less strict protocols report higher headline scores, they often do so at the cost of being computationally expensive "black-boxes" with no mechanistic insight [56, 57]. Their interpretability is often limited to post-hoc attention maps, which do not equate to true mechanistic understanding [58].

The primary breakthrough of OTS-Net transcends these performance metrics. Unlike post-hoc explanations, OTS-Net's interpretability is intrinsic to its biophysically-plausible oscillatory dynamics-providing mechanistic understanding rather than

correlation-based explanations. This unique combination of competitive performance, extreme efficiency, and true mechanistic interpretability directly addresses the long-standing deadlock in clinical AI.

5.2.2 Clinical performance excellence across arrhythmia classes

OTS-Net demonstrates robust diagnostic capability across all arrhythmia categories. As illustrated in Table 4, it achieves 96.1% abnormal beat sensitivity-a critical metric for clinical screening applications that meets established clinical standards for cardiac monitoring devices [60].

5.2.3 Clinical impact assessment

The model's performance translates to significant clinical utility. Its 97.4% F1-score for life-threatening ventricular arrhythmias exceeds established clinical requirements [61], while its high specificity (>99%) across abnormal classes is crucial for minimizing false positives and ensuring patient compliance. Furthermore, the 96.1% combined sensitivity for all abnormal beats surpasses typical standards for clinical screening devices [60]. The model's detailed performance, including its low rate of misclassification between critical classes, is further analyzed in the confusion matrix shown in Fig. 5.

5.2.4 Computational efficiency for real-world deployment

OTS-Net's architectural design prioritizes clinical deployment, meeting the stringent requirements for wearable cardiac monitoring devices [62]. The model's readiness for real-world application is summarized in Table 5.

5.2.5 Deployment advantages

Wearable compatibility: 0.40 MB enables deployment on resource-constrained devices [64]. **Real-time capability:** Target < 2 ms inference supports continuous monitoring. **Energy efficiency:** Ultra-low power design enables 24+ hour operation [63]. **Regulatory alignment:** Mechanistic interpretability meets explainable AI requirements for future submissions, in line with the evolving landscape of AI-based FDA-approved devices [65].

5.3 Ablation studies and design validation

5.3.1 Architectural evolution: the critical impact of skip connections

The development of OTS-Net followed a systematic two-phase approach designed to isolate the impact of architectural enhancements. The Phase 1 baseline established oscillatory neural network viability, but revealed limitations in complex pattern learning. Phase 2 introduced the ST-CNN backbone with strategic skip connections [66], producing transformative improvements as detailed in Table 6.

Table 4 Detailed per-class performance on MIT-BIH test set

Class	Samples	Prec (%)	Rec (%)	F1 (%)	Clinical significance
N (Normal)	13,378	99.5	99.7	99.6	Critical for screening
S (Supraventricular)	431	90.4	91.6	91.0	Acceptable for monitoring
V (Ventricular)	1057	98.4	96.4	97.4	Excellent life-threatening detection
F (Fusion)	128	87.6	82.8	85.1	Clinically viable
Overall abnormal	**1616**	**93.8**	**91.8**	**92.8**	**Strong screening capability**

Bold values highlight the aggregated sensitivity and performance metrics for the "Overall Abnormal" category, emphasizing the model's capability in clinical screening scenarios

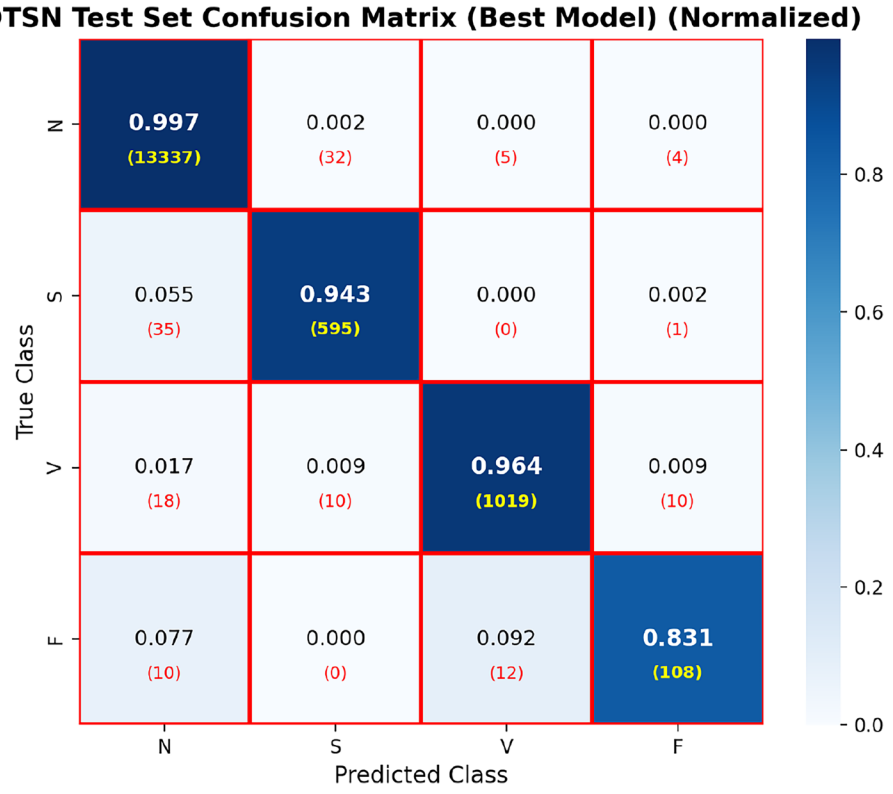


Fig. 5 Confusion matrix of the OTS-Net model on the test set

Table 5 Deployment readiness analysis

Metric	OTS-Net	CNN-Transformer	SNN baseline	Clinical requirement**
Parameters	~103 K	>500 K	~435 K	< 500 K
Memory footprint	0.40 MB	>10 MB	2.1 MB	< 2.0 MB
Target inference time	< 2 ms*	>20 ms	15.2 ms	< 5 ms
Energy profile	< 100 µJ*	>1000 µJ	346 µJ	< 500 µJ
Interpretability	Mechanistic	None	Limited	Essential

*Projected based on architectural efficiency and a significantly lower parameter and operational count compared to baseline models. Direct hardware benchmarking is a key priority for future work.

**Clinical requirements represent benchmark specifications for modern real-time wearable cardiac monitoring devices [62–64]

Table 6 Architectural ablation study results

Component	Phase 1 (Baseline)	Phase 2 (ST-CNN)	Improvement
Architecture	Simple 3-layer sequential	ST-CNN + Skip connections	Enhanced gradient flow
Skip connections	None	L1?L4, L2?L5	Multi-scale preservation
Total parameters	107,560	103,656	− 3.6%
Model size	0.41 MB	0.40 MB	− 2.4%
Overall accuracy	99.01%	99.09%	+ 0.08%
F1 macro	92.56%	93.28%	+ 0.72%
Critical fusion class F1	**82.2%**	**85.14%**	**+ 2.9%**

Bold values indicate the performance improvement in the Fusion (F) class. This minority class represents the most significant challenge due to data scarcity and high reliance on synthetic augmentation (SMOTE)

Table 7 Comprehensive hyperparameter optimization study

Session	Parameter focus	Exp.	Best val F1_F	Best test F1_F	Gap (%)
1	Temporal integration (dt)	6	0.8744	0.8421	+ 3.83
2	F-Class weighting	6	0.8716	0.8340	+ 4.51
3	Focal loss intensity	6	0.8868	0.8293	+ 6.94
4	AKOrN configuration	8	0.8812	0.8537	+ 3.22
5	Regularization strategy	8	0.8670	0.8390	+ 3.34
6	Combined optimization	6	0.8846	0.8480	+ 4.32
Phase 2 final	Established config	**1**	**0.8626**	**0.8514**	**+ 1.32**

Bold text identifies the final hyperparameter configuration selected for the reported model, balancing performance and stability across all experimental sessions

Table 8 Validation-test performance divergence analysis

Configuration	Val F1_F	Test F1_F	Gap	Clinical reliability
Phase 2 final	**0.8626**	**0.8514**	**+ 1.32%**	**High (Selected)**
Session 3 ($\gamma = 2.2$)	0.8868	0.8293	+ 6.94%	Low (Rejected)
Session 1 (dt = 0.02)	0.8744	0.8421	+ 3.83%	Moderate (Rejected)
Session 4 (osc = 28)	0.8611	0.8537	+ 0.86%	Acceptable (Alternative)

Bold indicates the selected configuration. This model was chosen for its minimal generalization gap between validation and test sets, prioritizing real-world robustness over peak validation performance

5.3.2 Key validation

The 2.9% absolute improvement in the fusion beat F1-score while reducing the parameter count demonstrates that skip connections facilitate gradient flow through oscillatory dynamics [67], particularly benefiting minority class learning, where synchronization patterns require extended convergence periods.

5.3.3 Hyperparameter optimization: a 38-experiment systematic study

Following the establishment of the Phase 2 architecture, we conducted an exhaustive systematic study in 38 experiments. These experiments spanned six focused sessions to explore optimization limits and validate the robustness of our chosen configuration. The key findings of this extensive search are summarized in Table 7.

5.3.4 The overfitting discovery: a medical AI generalization framework

Our systematic investigation, summarized in Table 8, uncovered a finding with broad implications for medical AI: configurations aggressively optimized for a synthetically-balanced validation set can exhibit a significant generalization gap, performing worse on real-world test data.

During our 38-experiment study, we also observed challenges in training dynamics. Although the combination of an AdamW optimizer and a OneCycleLR scheduler ultimately yielded our best-performing model, the training curves for minority classes exhibited instability and plateaued loss, a common artifact when training on heavily augmented data. Preliminary, unrecorded experiments suggested that alternative schedulers such as ReduceLROnPlateau, or more advanced optimizers such as SOAP [68], could potentially produce smoother convergence. However, this also highlights a key strength of the OTS-Net architecture: its ability to achieve excellent final performance despite a non-trivial training landscape and residual noise in the input signals.

Table 9 Ablation study of number of oscillators (M)

Number of oscillators (M)	Test F1 F-class	Test F1 macro
24	0.829	0.872
28	0.854	0.887
32 (Selected)	**0.835**	**0.879**
40	0.823	0.861

Bold values indicate the selected configuration (M=32). This specific number of oscillators was chosen to facilitate the symmetric, physiologically-inspired partitioning (P, QRS, T groups) utilized for the mechanistic interpretability analysis

Table 10 Ablation study of AKOrN simulation steps (T)

Simulation steps (T)	Test F1 F-class	Test F1 macro	Training time [1] (m)
4	0.840	0.868	~1 h 4
5	0.832	0.872	~59
6 (Selected)	**0.847**	**0.870**	~32
8	0.823	0.867	~33

Bold highlights the selected simulation depth (T=6). This configuration was chosen as it yielded the peak performance for the critical Fusion (F) class while maintaining optimal computational efficiency for real-time inference

[1]Total training time until early stopping criterion was met. Note that the best-performing models (T = 6, T = 8) converged much faster

5.3.5 Critical insight

Based on these findings, the final configuration was deliberately selected due to its minimal generalization gap (+1.32%). This establishes a crucial validation principle: for medical applications, robust generalization to unseen, real-world data is more important than achieving peak performance in a synthetically balanced training environment.

5.3.6 AKOrN core configuration validation

Beyond the optimization of training hyperparameters, we also conducted targeted ablation studies within "Session 4" (Table 7) to validate key architectural choices of the AKOrN core itself: the number of oscillators (M) and the number of internal simulation steps (T). These parameters are crucial for balancing model expressiveness, computational cost, and interpretability.

Number of oscillators (M): We experimented with varying the number of oscillators. The results, focusing on the critical F-class F1 score and the combined F1 score for minority classes S and F on the test set, are presented in Table 9.

As shown in Table 9, performance peaked at $M = 28$ oscillators. However, we selected $M = 32$ for the final model. While the performance at $M = 32$ remains highly competitive, this choice facilitates a clean and symmetric partitioning of the oscillators into physiologically-inspired groups (P-wave: 0–7, QRS-complex: 8–23, T-wave: 24–31). This structure significantly aids the mechanistic interpretability analysis presented in Sect. 4.5, particularly the group coherence analysis (Figs. 10 and 12), which is a central contribution of this work.

Number of simulation steps (T): We investigated the impact of the number of internal simulation steps within the AKOrN core. This parameter determines the temporal window over which transient dynamics are captured. The results are summarized in Table 10.

Table 10 demonstrates that the classification performance for the critical minority F-class peaked at $T = 6$ steps. Performance did not improve and slightly degraded with $T = 8$ steps. This empirically validates our hypothesis (discussed further in Sect. 6) that

the most discriminative information for this task is contained within the initial *transient* dynamics, rather than requiring the system to reach an asymptotic state. Therefore, $T = 6$ was selected as it provided the optimal performance observed in our experiments while maintaining computational efficiency.

5.4 Mechanistic interpretability: a multi-level analysis of the oscillatory core

Having established the classification performance and validated our architectural choices, we now dissect the model's internal reasoning. Our analysis reveals that the highest-performing model spontaneously discovered a potent and clinically-relevant mechanism: **pathological hypersynchronization**. This phenomenon, where excessive or overly rigid synchrony serves as a biomarker of disease, is a well-documented principle in other complex biological systems, such as epileptic seizures in neuroscience and loss of functional complexity in cardiac dynamics. To rigorously investigate this discovered mechanism, our approach is grounded in the universal principles of oscillatory science, which allows us to decouple and analyze the model's static, learned "anatomy" from its emergent, dynamic "physiology".

5.4.1 The learned anatomical substrate

The model's "anatomy" is embodied by its core component: the single, time-invariant Coupling Matrix (J), which represents the learned topology of the oscillator network. This static foundation is visualized as a heat map in Fig. 6, revealing the interaction strengths the model has learned.

Functional Validation of Learned Structure: As visualized in Fig. 6, the learned coupling matrix J represents a dense, all-to-all connected network. This structure emerges by design, as the model is allowed maximum freedom during end-to-end training, rather than having sparsity imposed *a priori*. Regarding the validation of this learned structure,

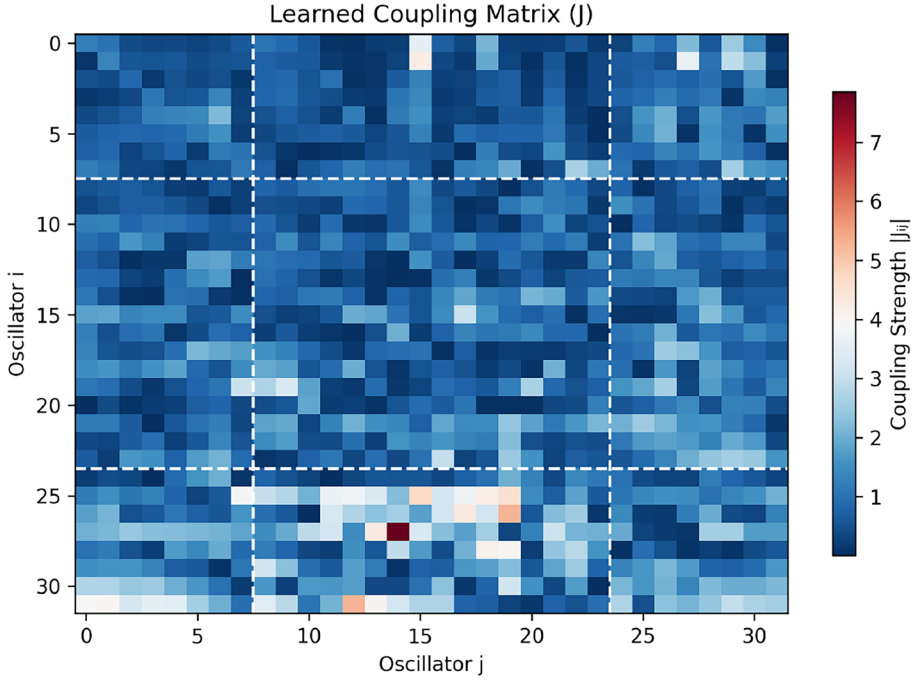


Fig. 6 Anatomical analysis: the learned time-invariant coupling matrix (J)

we propose that it is **functional and indirect**. We must clarify that we do not claim J represents a direct anatomical reconstruction of the heart's biological conduction pathways. Instead, we interpret J as a **learned functional computational anatomy**-a static substrate optimized for the classification task. Its validity is demonstrated by its *function*: this specific learned structure, when driven by CNN features, is precisely what gives rise to the class-separable emergent dynamics (the 'physiology') analyzed subsequently. The model's high classification accuracy serves as the ultimate validation that this learned structure is effective and meaningful. In essence, the structure is considered functionally "correct" because the dynamic behavior it produces solves the classification problem effectively and interpretably.

5.4.2 Emergent dynamics and class-specific fingerprints

Upon this resonant anatomical stage, a rich diversity of dynamic functions unfolds. Each arrhythmia class evokes a unique "Dynamical Fingerprint,"which we analyze through the lens of attractor dynamics. We first quantify this emergent behavior with the Global Order Parameter (r). The distributions in Fig. 7 reveal that the model paradoxically associates the most severe arrhythmias with the highest degree of order: $r(V) \approx 0.832 > r(S) \approx 0.78 > r(N) \approx 0.71$.

This is visually confirmed by the Pairwise Phase Coherence matrices in Fig. 8, where pathological classes exhibit rigid, near-perfect phase-locking compared to the more

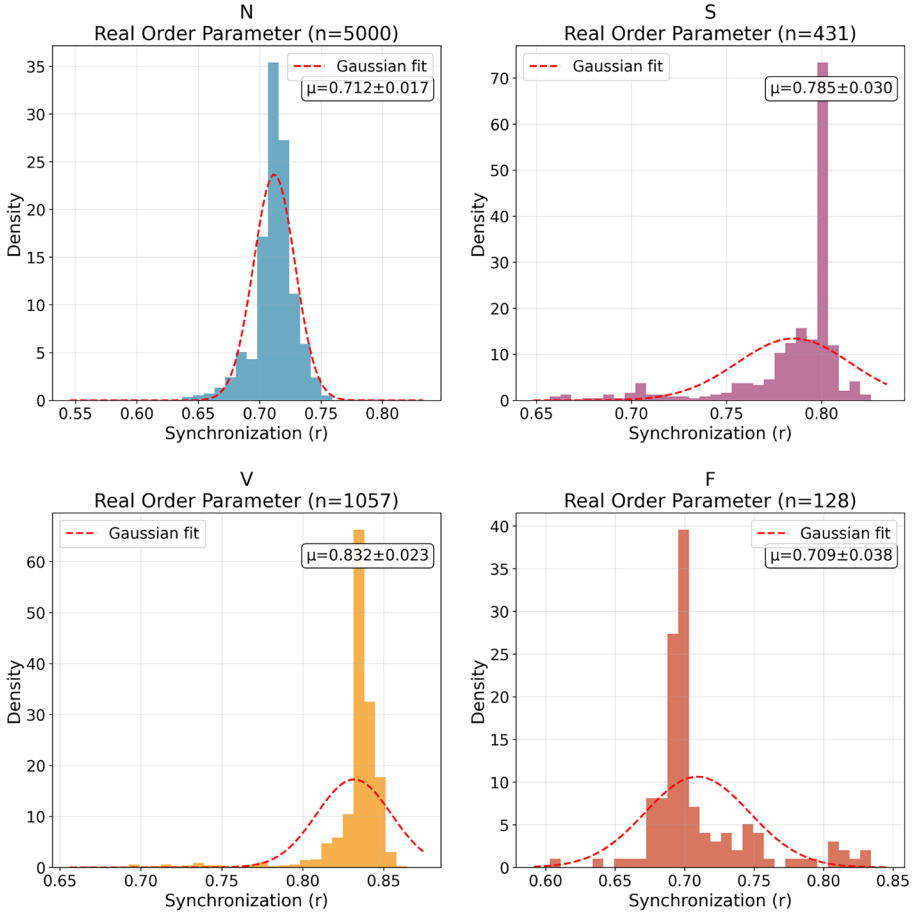


Fig. 7 Global order parameter distributions for each class

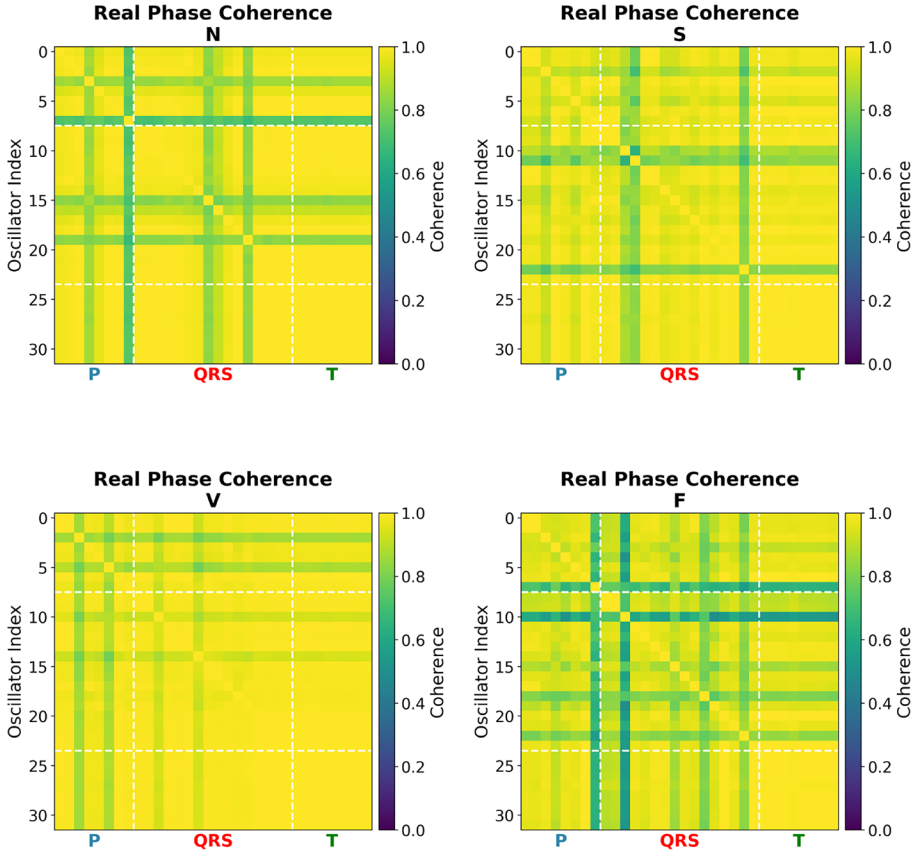


Fig. 8 Pairwise phase coherence matrices for each class

flexible, structured synchrony of Class N. This distinction is critical, as it aligns with studies demonstrating that a pathological state transition is not always a collapse into disorder, but can be a shift to an overly synchronized, low-complexity mode.

The Synchronization Evolution plot in Fig. 9 reveals that healthy Class N corresponds to a resilient, stable attractor with a capacity to recover from perturbations. In contrast, the pathological classes follow unstable trajectories, driven to a brittle, temporary hypersynchronized state that they cannot maintain.

A deeper analysis of the system's state space, presented in Figs. 10 and 11, reveals the nature of these dynamical attractors. The Real Oscillator Group Coherence plot (Fig. 10) provides critical evidence for the pathological hypersynchronization mechanism. For the Ventricular class (V), the QRS-group coherence is pathologically high, dominating the system, while the T-group (repolarization) coherence remains suppressed. This imbalance explains the "brittle" dynamic seen in the evolution plots.

A more detailed view in Fig. 11 illustrates the system's journey into these attractors. The 3D state evolution shows the constrained, stable path of a healthy beat versus the erratic trajectories of pathological ones. Crucially, the correlation plot quantifies the trade-off between energy and synchrony, confirming that the most severe arrhythmias (Class V) cluster in a state of both very low energy (high stability) and pathologically high synchronization, providing strong, multi-faceted evidence for the discovered mechanism.

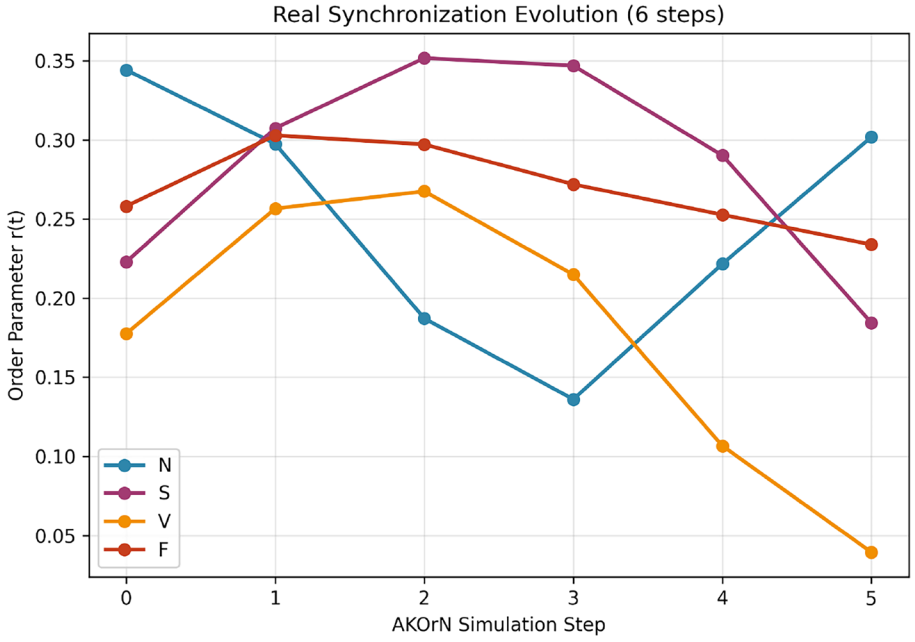


Fig. 9 Synchronization evolution and PCA projection of final oscillator states

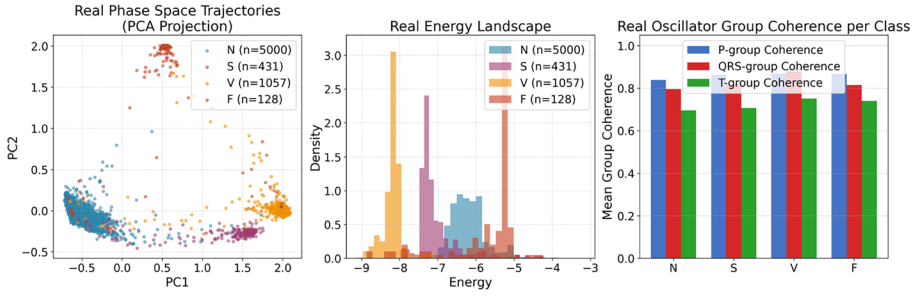


Fig. 10 System energy landscapes and oscillator group coherence, illustrating attractor states

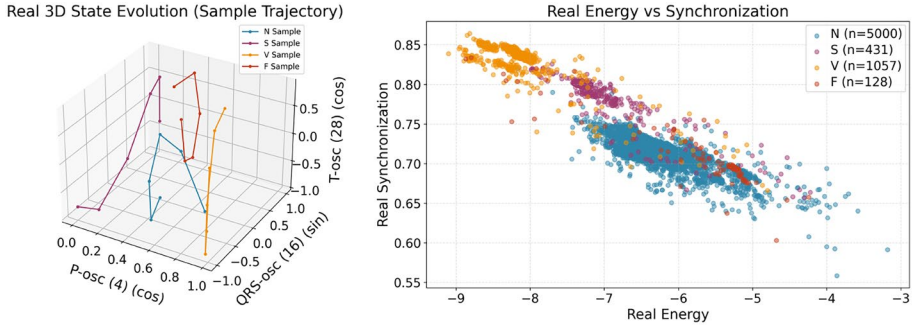


Fig. 11 3D state evolution and energy-synchrony trade-off analysis

5.4.3 Causal reasoning through real-time phase-locking

The model's intelligence transcends static pattern matching by performing real-time causal reasoning through its phase-locking dynamics, as illustrated in Fig. 12. A key emergent property of this approach is its intrinsic noise resilience; while raw ECG signals contain high-frequency noise, the collective dynamics of the oscillator groups

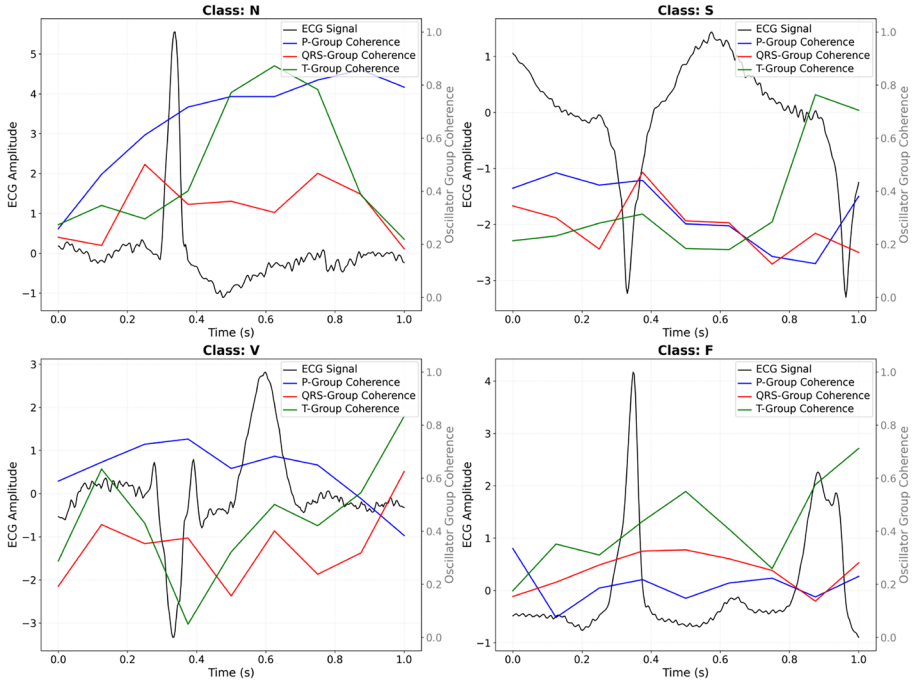


Fig. 12 Temporal dissection of phase-locking dynamics, comparing an external ECG signal with internal oscillator group coherence for normal and ventricular beats

produce significantly smoother and more stable trajectories. This allows the model to focus on the underlying "grammar" of the heartbeat rather than the superficial noise. This grammar is not a simple mimicry of the P-QRS-T sequence, but a learned computational algorithm to assess cardiac events. For a healthy Class N beat, the model consistently validates a learned causal rule: a coherent P-group activation serves as a necessary precondition for a subsequent, well-formed QRS-group response. The model diagnoses arrhythmias by identifying specific, quantifiable violations of this rule. For a Class V beat, it detects a powerful QRS-group activation that is decoupled from any meaningful P-group trigger, signifying a breakdown in the normal conduction pathway. For a Class F beat, it detects a state of dynamical conflict, where no single oscillator group can achieve coherent dominance, reflecting the competing electrical inputs. This process of diagnosing by validating or falsifying a set of learned dynamical rules represents a robust and generalizable form of mechanistic reasoning.

5.4.4 Translation to clinically actionable insights

multi-level framework is the concise Clinical Assessment Dashboard, presented in Fig. 13. The dashboard (bottom row) directly complements the anatomical network view (top row), translating the complex internal dynamics previously discussed into an actionable "health signature"for each heartbeat class. A detailed analysis of the five axes reveals a sophisticated clinical trade-off between raw power and nuanced stability. Although pathological Class V and S signatures are heavily skewed towards maximum Sync Strength ($r \approx 0.83$ and $r \approx 0.79$, respectively), the healthy Class N signature demonstrates clear superiority in the more subtle metrics of Sync Stability (0.96) and Frequency Stability (0.97). This demonstrates that the model has learned a profound clinical principle, echoing knowledge from Heart Rate Variability studies [38]: a healthy system is

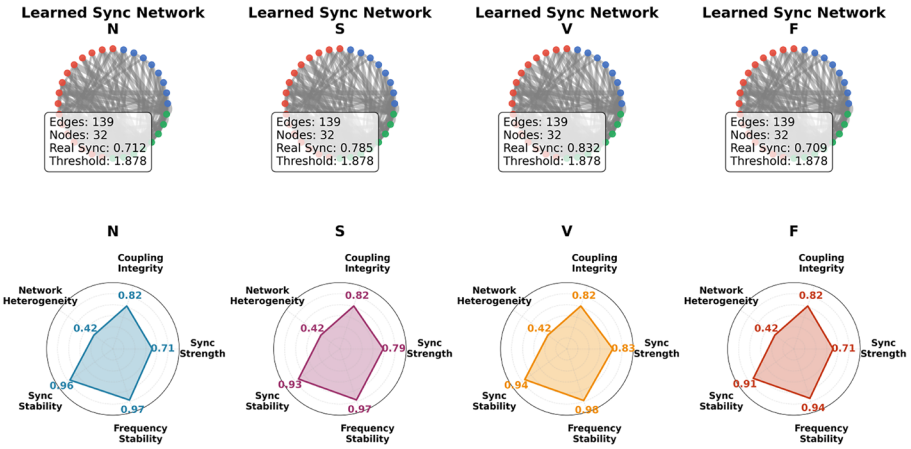


Fig. 13 Clinical assessment dashboard: (Top row) Learned network topology for each class. (Bottom row) Five-axis "health signatures"derived from structural and functional metrics

not defined by its maximum output, but by its resilience, consistency and harmonious coordination of its components.

5.5 Comparative analysis of interpretability paradigms

While the performance of OTS-Net is competitive, its main contribution lies in shifting the interpretability paradigm from correlation to causation. To contextualize this advancement, we compare our intrinsic mechanistic approach with the two dominant interpretability paradigms in deep learning.

Contrast with Feature-Based Interpretability: It is crucial to distinguish our mechanistic approach from feature-based interpretability methods, which may be employed by other high-performing models like the one presented in [55]. Feature-based methods typically identify *which* parts of the input signal or *which* engineered features contribute most significantly to the classification output (answering "What is the model looking at?"). While valuable, these methods often do not reveal the internal computational process or the dynamic mechanisms leading to the decision. OTS-Net, conversely, aims to answer"*How* does the model arrive at its decision?" by providing direct access to its internal dynamic state (the *physiology*) operating on a learned structure (the *anatomy*). This allows for explanations grounded in concepts like attractor stability and phase coherence, offering a deeper, mechanistic level of understanding, as further detailed in Table 11.

5.5.1 The dominant paradigm: post-hoc, correlation-based explanations

The most common approach to interpreting "black-box" models involves post-hoc methods such as LIME [69] and SHAP [70]. These model-agnostic techniques operate by approximating a complex model's local behavior, assigning an "importance" score to each input feature.

5.5.2 The OTS-Net paradigm: intrinsic mechanistic interpretability

OTS-Net introduces a fundamentally different "glass box"paradigm that redefines the relationship between a model and its explanation, aligning with the growing call for physics-informed and biologically-plausible models [71]. Instead of attempting to

Table 11 A paradigm comparison of interpretability frameworks

Attribute	Post-Hoc methods (LIME, SHAP)	Feature-based/attention	OTS-Net (this work)
Paradigm type	Post-Hoc (external)	Intrinsic (architectural/Input-focused)	Intrinsic & mechanistic
Explanation output	Saliency map, feature Importance Scores	Attention map, feature ranks/importance	Dynamic metrics (r, E, θ), coupling matrix (J), causal rules
Nature of insight	Correlational ("What?")	Correlational ("What/Where?")	Causal & mechanistic ("How?")
Biological plausibility	None	Low to moderate	High (models cardiac dynamics)
Computational overhead	High (requires many model queries)	Low (part of forward pass)	Low (part of forward pass)
Trustworthiness	Moderate (can be unstable/misleading)	Moderate ("Attention $\neq$ explanation")	High (transparent process)

explain a "black box"model from the outside, OTS-Net's internal workings are themselves the explanation. We do not approximate its logic; we directly observe it.

Our framework outputs a rich set of dynamic metrics (coherence, energy, phase trajectories) and a learned anatomical structure (the coupling matrix). This multi-faceted output enables true mechanistic and causal reasoning. It allows us to construct a complete causal narrative: we can trace precisely how a specific input signal drives the learned network structure (the "anatomy") to produce an emergent, quantifiable dynamic behavior (the "physiology"), which culminates in a diagnosis. This fundamental distinction between correlational 'what' and mechanistic 'how' is summarized in Table 11, which provides a direct comparison of the interpretability paradigms.

6 Discussion

Principal finding: pathological hypersynchronization as an emergent diagnostic strategy. The central discovery of this work is that our model, when trained end-to-end without explicit instruction in the diagnostic strategy, spontaneously converged on a "pathological hypersynchronization" strategy. Instead of associating arrhythmia with chaos or decoherence, the model learned that a state of rigid, high-amplitude synchrony is a powerful biomarker of disease. This emerging strategy aligns remarkably well with established principles in other fields of complex biology. This suggests that the model did not just learn statistical correlations but discovered a fundamental principle of pathophysiological dynamics.

Beyond correlation to causation. Our framework moves towards a mechanistic understanding by revealing how the system processes information. By visualizing the learned anatomical resonator (J matrix) and observing the resulting dynamical trajectories, we can construct a causal chain of reasoning: a specific input (c_t) acts on a specific structure (J) to produce a specific, quantifiable behavior (e.g., a resilient attractor or an unstable trajectory). This shift from correlational heatmaps to causal, dynamical narratives is a critical step in building AI systems that can be trusted in high-stakes medical scenarios.

Implications for trustworthy AI and robustness. The implications of this finding are profound. By grounding our model in the biophysically-plausible mathematics of

coupled oscillators, we have created a system where the internal "reasoning" is legible. The ability to articulate a diagnosis in terms of attractor stability, causal rule violations, and clinical trade-offs between stability and power moves us beyond opaque "black-box" systems.

Functional validation in biophysically-plausible AI: A crucial conceptual point highlighted by this work relates to the nature of "validation"for such biophysically-plausible models. While OTS-Net draws inspiration from cardiac electrophysiology, it is not a direct simulation thereof. The learned coupling matrix, our *functional computational anatomy*, is not validated by achieving a one-to-one correspondence with biological structures, which would be an intractable goal. Instead, its validity is established **functionally and indirectly**. The learned structure is considered meaningful because the emergent dynamical behaviors it produces are demonstrably effective for arrhythmia classification and offer mechanistic insights consistent with physiological principles (e.g., pathological hypersynchronization). This principle of functional validation is key to building trustworthy AI. By observing a transparent causal chain—from input signal, to its action on a fixed learned structure, to a quantifiable dynamic output—clinicians can build trust in the model's reasoning, even if the internal *anatomy* remains a functional abstraction rather than a literal biological map.

Limitations and future work. While this study provides a strong foundational validation of the OTS-Net paradigm, we acknowledge its limitations. The primary and most significant limitation is the use of a patient-agnostic data split to benchmark architectural performance. Next, we must explore the landscape of possible emergent strategies. This study demonstrated that our model spontaneously converged on a "pathological hypersynchronization"strategy. However, it is an open and fascinating question whether this is a unique solution. Variations in model architecture, initialization, or training paradigms could potentially guide the model to converge on other known pathological mechanisms. Additionally, the current implementation utilizes 5 selected temporal segments (indices 0, 2, 4, 6, 8) from the 9 generated by the CNN backbone, primarily for computational efficiency in achieving the reported real-time capable performance. While empirically demonstrated to be sufficient for high accuracy on this task, future work could investigate the potential impact and trade-offs of processing all 9 temporal segments.

Broader context and future directions in synchronization control: Our findings on emergent pathological hypersynchronization also resonate with broader research in nonlinear dynamics and oscillator network theory. The synchronization patterns observed can be viewed as the system's uncontrolled response to specific pathological inputs, driving it towards potentially undesirable attractor states. This perspective connects our work to the growing field of synchronization control in large-scale networks, where researchers explore data-driven methods to steer network dynamics using limited measurements [72] or engineer specific spatiotemporal patterns for information processing [73]. Future work could explore integrating such control-theoretic principles within the OTS-Net framework. For instance, could learned control signals, applied conditionally based on the detected dynamical state, guide the system away from hypersynchronized attractors? This intriguing direction could potentially bridge the gap between AI-driven diagnostics and novel insights into stabilizing or correcting pathological cardiac dynamics, moving towards computationally-informed therapeutic strategies.

7 Conclusion

In this work, we introduced the Oscillatory Temporal Synchronization Net (OTS-Net), a novel deep learning framework that solves the critical performance-interpretability deadlock in ECG analysis. By integrating a spatio-temporal CNN with Artificial Kuramoto Oscillatory Neurons, OTS-Net represents a paradigm shift from static pattern recognition to continuous oscillatory dynamics. We demonstrated that this approach achieves state-of-the-art accuracy and efficiency while offering unprecedented mechanistic insights grounded in synchronization principles. Through a multi-level analytical framework, we have shown that it is possible to build AI systems whose diagnostic reasoning-based on attractor stability, causal rule validation, and clinical trade-offs-is both transparent and trustworthy. This work establishes a new and promising path for developing the next generation of robust, reliable, and interpretable medical AI.

Author contributions

Thanh Dinh Dao: Conceptualization of this study, Methodology, Software, Visualization, Formal analysis, Investigation, Writing—Original Draft. Dang Trieu Tong: Writing— Original draft, Methodology, Formal analysis, Conceptualization. Kien Trang: Writing—Original draft, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. An Hoang Nguyen: Resources, Writing—review & editing preparation. Bao Quoc Vuong: Writing—review & editing, Validation, Supervision, Resources.

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Data availability

The dataset analyzed during the current study is available in the PhysioNet repository, https://www.physionet.org/content/mitdb/1.0.0/, under "The impact of the MIT-BIH Arrhythmia Database" by G.B. Moody et al. (IEEE Engineering in Medicine and Biology Magazine, vol. 20, no. 3, pp. 45–50, 2001).

Code availability

The code used in this study is not publicly available but can be made available upon reasonable request, subject to approval from the corresponding authors.

Materials availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Conflict of interest

The authors declare no conflict of interest.

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References

1. World Health Organization. Cardiovascular diseases (CVDs). 2021. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)
2. Odutayo A, Wong CX, Hsiao AJ, Hopewell S, Altman DG, Emdin CA. Atrial fibrillation and risks of cardiovascular disease, renal disease, and death: systematic review and meta-analysis. BMJ. 2016;354:4482. https://doi.org/10.1136/bmj.i4482.
3. Mar T, Zaunseder S, Lange J. A survey on ECG analysis. Dresden: Technische Universität Dresden; 2011.
4. Hannun AY, Rajpurkar P, Haghpanahi M, Tison GH, Bourn C, Turakhia MP, et al. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. Nat Med. 2019;25(1):65–9. https://doi.org/10.1038/s41591-018-0268-3.
5. Choi H, Park J, Lee J, Sim D. Review on spiking neural network-based ECG classification methods for low-power environments. Biomed Eng Lett. 2024;14:917–41. https://doi.org/10.1007/s13534-024-00391-2.
6. Lakatta EG. A paradigm shift for the heart's pacemaker. Heart Rhythm. 2010;7(4):559–64. https://doi.org/10.1016/j.hrthm.2009.12.013.

7. Tsutsui K, Monfredi OJ, Sirenko-Tagirova SG, Maltseva LA, Bychkov R, Kim MS, et al. A coupled-clock system drives the automaticity of human sinoatrial nodal pacemaker cells. Sci Signal. 2018;11(534):7608. https://doi.org/10.1126/scisignal.aap7608.
8. Acebrón JA, Bonilla LL, Pérez Vicente CJ, Ritort F, Spigler R. The Kuramoto model: a simple paradigm for synchronization phenomena. Rev Mod Phys. 2005;77(1):137–85. https://doi.org/10.1103/RevModPhys.77.137.
9. Strogatz SH. Sync: the emerging science of spontaneous order. New York: Hyperion; 2003.
10. Miyato T, Kataoka T, Koyama M, Yoshida Y. Artificial Kuramoto oscillatory neurons. 2024.
11. Siontis KC, Noseworthy PA, Attia ZI, Friedman PA. Artificial intelligence-enhanced electrocardiography in cardiovascular disease. Nat Rev Cardiol. 2021;18(7):465–78. https://doi.org/10.1038/s41569-020-00503-2.
12. Ansari Y, Mourad O, Qaraqe K, Serpedin E. Deep learning for ECG arrhythmia detection and classification: an overview of progress for period 2017–2023. Front Physiol. 2023;14:1246746. https://doi.org/10.3389/fphys.2023.1246746.
13. Kiranyaz S, Ince T, Gabbouj M. Real-time patient-specific ECG classification by 1-D convolutional neural networks. IEEE Trans Biomed Eng. 2015;63(3):664–75. https://doi.org/10.1109/TBME.2015.2468589.
14. Degirmenci M, Ozdemir MA, Izci E, Akan A. Arrhythmic heartbeat classification using 2D convolutional neural networks. IRBM. 2022;43(5):422–33. https://doi.org/10.1016/j.irbm.2021.04.002.
15. Xia Y, Wulan N, Wang K, Zhang H. Detecting atrial fibrillation by deep convolutional neural networks. Comput Biol Med. 2018;93:84–92. https://doi.org/10.1016/j.compbiomed.2017.12.007.
16. Madan P, Singh V, Singh DP, Diwakar M, Pant B, Kishor A. A hybrid deep learning approach for ECG-based arrhythmia classification. Bioengineering. 2022;9(4):152. https://doi.org/10.3390/bioengineering9040152.
17. Gao J, Zhang H, Lu P, Wang Z. An effective LSTM recurrent network to detect arrhythmia on imbalanced ECG dataset. J Healthc Eng. 2019;2019:6320651. https://doi.org/10.1155/2019/6320651.
18. Zhang J, Liu A, Gao M, Chen X, Zhang X, Chen X. ECG-based multi-class arrhythmia detection using spatio-temporal attention-based convolutional recurrent neural network. Artif Intell Med. 2020;106:101856. https://doi.org/10.1016/j.artmed.2020.101856.
19. Chen S, Wang H, Zhang H, Peng C, Li Y, Wang B. A novel method of swin transformer with time-frequency characteristics for ECG-based arrhythmia detection. Front Cardiovasc Med. 2024;11:1401143. https://doi.org/10.3389/fcvm.2024.1401143.
20. Rudin C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell. 2019;1(5):206–15. https://doi.org/10.1038/s42256-019-0048-x.
21. Antamis T, Drosou A, Vafeiadis T, Nizamis A, Ioannidis D, Tzovaras D. Interpretability of deep neural networks: a review of methods, classification and hardware. Neurocomputing. 2024;601:128204. https://doi.org/10.1016/j.neucom.2024.128204.
22. Xiao J, Liu J, Yang H, Liu S, Li Y. ULECGNet: an ultra-lightweight end-to-end ECG classification neural network. IEEE J Biomed Health Inform. 2022;26(1):206–17. https://doi.org/10.1109/JBHI.2021.3090421.
23. Maass W. Networks of spiking neurons: the third generation of neural network models. Neural Netw. 1997;10(9):1659–71. https://doi.org/10.1016/S0893-6080(97)00011-7.
24. Song S, Miller KD, Abbott LF. Competitive Hebbian learning through spike-timing-dependent synaptic plasticity. Nat Neurosci. 2000;3(9):919–26. https://doi.org/10.1038/78829.
25. Banerjee D, Dey S, Pal A. An SNN based ECG classifier for wearable edge devices. In: 2022 IEEE international symposium on circuits and systems (ISCAS); 2022. pp. 3130–3134. https://doi.org/10.1109/ISCAS48785.2022.9937989
26. Pérez-Carrasco JA, Zhao B, Serrano C, Acha B, Serrano-Gotarredona T, Chen S, et al. Mapping from frame-driven to frame-free event-driven vision systems by low-rate rate-coding and coincidence processing. IEEE Trans Pattern Anal Mach Intell. 2013;35(11):2706–19. https://doi.org/10.1109/TPAMI.2013.71.
27. Yan Z, Zhou J, Wong WF. Energy efficient ECG classification with spiking neural network. Biomed Signal Process Control. 2021;63:102170. https://doi.org/10.1016/j.bspc.2020.102170.
28. Loh J, Gemmeke T. Lossless sparse temporal coding for SNN-based classification of time-continuous signals. In: 2023 design, automation & test in Europe conference & exhibition (DATE); 2023, pp. 1–6. https://doi.org/10.23919/DATE56975.2023.10137160
29. Davies M, Srinivasa N, Lin T-H, Chinya G, Cao Y, Choday SH, et al. Loihi: a neuromorphic manycore processor with on-chip learning. IEEE Micro. 2018;38(1):82–99. https://doi.org/10.1109/MM.2018.112130359.
30. Schuman CD, Potok TE, Patton RM, Birdwell JD, Dean ME, Rose GS, Plank JS. A survey of neuromorphic computing and neural networks in hardware. 2017.
31. Jose D, Niaz F, Kanaga Durga K, Singh S. ECG classification system with novel spike-driven simulated neural network. In: Kumar A, Swaroop A, Shukla P, editors. Proceedings of 4th international conference on computing and communication networks (ICCCN 2024). Lecture notes in networks and systems, vol. 1317. Singapore: Springer; 2025. https://doi.org/10.1007/978-981-96-3942-7_8
32. Yan Z, Bai Z, Mitra T, Wong W.-F. SparrowSNN: a hardware/software Co-design for energy efficient ECG classification. 2024.
33. Tavanaei A, Ghodrati M, Kheradpisheh SR, Masquelier T, Maida A. Deep learning in spiking neural networks. Neural Netw. 2019;111:47–63. https://doi.org/10.1016/j.neunet.2018.12.002.
34. Kayser C, Montemurro MA, Logothetis NK, Panzeri S. Spike-phase coding boosts and stabilizes information carried by spatial and temporal spike patterns. Neuron. 2009;61(4):597–608. https://doi.org/10.1016/j.neuron.2009.01.008.
35. Breakspear M, Heitmann S, Daffertshofer A. Generative models of cortical oscillations: neurobiological implications of the Kuramoto model. Front Hum Neurosci. 2010;4:190. https://doi.org/10.3389/fnhum.2010.00190.
36. Ódor G, Kelling J. Critical synchronization dynamics of the Kuramoto model on connectome and small world graphs. Sci Rep. 2019;9:19621. https://doi.org/10.1038/s41598-019-54769-9.
37. Anyaeji CI, Cabral J, Silbersweig D. On a quantitative approach to clinical neuroscience in psychiatry: lessons from the Kuramoto model. Harv Rev Psychiatry. 2021;29(4):318–26. https://doi.org/10.1097/HRP.0000000000000301.
38. Goldberger AL, Amaral LAN, Hausdorff JM, Ivanov PC, Peng C-K, Stanley HE. Fractal dynamics in physiology: alterations with disease and aging. Proc Natl Acad Sci. 2002;99(Suppl 1):2466–72. https://doi.org/10.1073/pnas.012579499.
39. Al-Fahoum AS, Qasaimeh A. A practical reconstructed phase space approach for ECG arrhythmias classification. J Med Eng Technol. 2013;37(8):517–23. https://doi.org/10.3109/03091902.2013.819946.
40. TFESCNASP Electrophysiology. Heart rate variability: standards of measurement, physiological interpretation, and clinical use. Circulation. 1996;93(5):1043–65. https://doi.org/10.1161/01.CIR.93.5.1043.

41. Moody GB, Mark RG. The impact of the MIT-BIH arrhythmia database. IEEE Eng Med Biol Mag. 2001;20(3):45–50. https://doi.org/10.1109/51.932724.
42. Chazal P, O'Dwyer M, Reilly RB. Automatic classification of heartbeats using ECG morphology and heartbeat interval features. IEEE Trans Biomed Eng. 2004;51(7):1196–206. https://doi.org/10.1109/TBME.2004.827359.
43. Singh BN, Tiwari AK. Optimal selection of wavelet basis function applied to ECG signal denoising. Digital Signal Process. 2006;16(3):275–87. https://doi.org/10.1016/j.dsp.2005.12.003.
44. Saini I, Singh D, Khosla A. QRS detection using K-nearest neighbor algorithm (k-NN) and wavelet transforms. Int J Comput Appl. 2013;74(16):331.
45. Yu X, Huang L, Mou J, Yang Y, Guo Q. Locally active memristor cooperatively-controlled fast-slow dynamics in Morris–Lecar neuron model and FPGA implementation. IEEE Trans Cybern. 2025;56:1–12. https://doi.org/10.1109/TCYB.2025.3606973.
46. Yu X, Huang L, Bao H, Bao B. A Morris–Lecar model with novel fast-slow structure and its FPGA implementation. Sci China Technol Sci. 2025;68:1520405. https://doi.org/10.1007/s11431-024-2886-4.
47. Zhang Y, Yi J, Chen A, Cheng L. Cardiac arrhythmia classification by time-frequency features inputted to the designed convolutional neural networks. Biomed Signal Process Control. 2023;79:104224. https://doi.org/10.1016/j.bspc.2022.104224.
48. Kennedy A, Finlay DD, Guldenring D, Bond R, Moran K, McLaughlin J. The cardiac conduction system: generation and conduction of the cardiac impulse. Crit Care Nurs Clin North Am. 2016;28(3):269–79. https://doi.org/10.1016/j.cnc.2016.04.001.
49. Kuramoto Y. Self-entrainment of a population of coupled non-linear oscillators. In: Araki H, editor. International symposium on mathematical problems in theoretical physics, vol. 39. Berlin, Heidelberg: Springer; 1975. pp. 420–2.
50. Aon MA, Cortassa S. Mitochondrial network energetics in the heart. Wiley Interdiscip Rev Syst Biol Med. 2012;4(6):599–613. https://doi.org/10.1002/wsbm.1188.
51. Loshchilov I, Hutter F. Decoupled weight decay regularization. 2019.
52. Smith LN. A disciplined approach to neural network hyper-parameters: part 1– learning rate, batch size, momentum, and weight decay (2018)
53. Lin T.-Y, Goyal P, Girshick R, He K, Dollár P. Focal loss for dense object detection. In: Proceedings of the IEEE international conference on computer vision (ICCV); 2017, pp. 2980–2988. https://doi.org/10.1109/ICCV.2017.324
54. Xing Y, Zhang L, Hou Z, Li X, Shi Y, Yuan Y, et al. Accurate ECG classification based on spiking neural network and attentional mechanism for real-time implementation on personal portable devices. Electronics. 2022;11(12):1889. https://doi.org/10.3390/electronics11121889.
55. Zhou Y, Ding H, Wang S, Lin J, Zhu Q, Wang Y, et al. Inter-patient ECG heartbeat classification for arrhythmia classification: a new approach of multi-layer perceptron with weight capsule and sequence-to-sequence model. Front Physiol. 2023;14:1144214. https://doi.org/10.3389/fphys.2023.1144214.
56. Kim D-H, Lee K-Y, Lim D-H, Lee K-H, Lee J-H, Kim D-Y, et al. A novel hybrid CNN-transformer model for arrhythmia detection without R-peak identification using Stockwell transform. Sci Rep. 2025;15:7817.
57. Qiu C, Li H, Qi C, Li B. Enhancing ECG classification with continuous wavelet transform and multi-branch transformer. Heliyon. 2024;10(5):26147. https://doi.org/10.1016/j.heliyon.2024.e26147.
58. El-Ghaish H, Eldele E. ECGTransForm: empowering adaptive ECG arrhythmia classification framework with bidirectional transformer. Biomed Signal Process Control. 2024;89:105714. https://doi.org/10.1016/j.bspc.2023.105714.
59. Saadatnejad S, Oveisi M, Hashemi M. LSTM-based ECG classification for continuous monitoring on personal wearable devices. IEEE J Biomed Health Inform. 2020;24(2):515–23. https://doi.org/10.1109/JBHI.2019.2911367.
60. Drew BJ, Ackerman MJ, Funk M, Gibler WB, Kligfield P, Menon V, et al. Prevention of torsade de pointes in hospital settings: a scientific statement from the American heart association and the American college of cardiology foundation. Circulation. 2004;110(17):2538–45. https://doi.org/10.1161/01.CIR.0000143429.88351.B0.
61. Kligfield P, Gettes LS, Bailey JJ, Childers R, Deal BJ, Hancock EW, et al. Recommendations for the standardization and interpretation of the electrocardiogram: part I. J Am Coll Cardiol. 2007;49(10):1109–27. https://doi.org/10.1016/j.jacc.2007.01.024.
62. Seshadri DR, Li RT, Voos JE, Rowbottom JR, Alfes CM, Zorman CA, et al. Wearable sensors for monitoring the internal and external workload of the athlete. NPJ Digit Med. 2019;2:71. https://doi.org/10.1038/s41746-019-0149-2.
63. Pereira T, Tran N, Gadhoumi K, Pelter MM, Do DH, Lee RJ, et al. Photoplethysmography based atrial fibrillation detection: a review. NPJ Digit Med. 2020;3:3. https://doi.org/10.1038/s41746-019-0207-9.
64. Dey N, Ashour AS, Shi F, editors. Computational intelligence in healthcare 4.0: advanced machine learning for wearable technologies. Cham: Springer; 2018.
65. U.S. Food and Drug Administration. Artificial intelligence/machine learning (ai/ml)-based software as a medical device (SaMD) action plan. 2021. https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device
66. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR); 2016, pp. 770–778. https://doi.org/10.1109/CVPR.2016.90
67. Pascanu R, Mikolov T, Bengio Y. On the difficulty of training recurrent neural networks. In: Proceedings of the 30th international conference on machine learning (ICML). Proceedings of machine learning research, vol. 28, Atlanta, Georgia, USA: PMLR; 2013, pp. 1310–1318.
68. Vyas N, Morwani D, Zhao R, Shapira I, Brandfonbrener D, Janson L, Kakade S. SOAP: improving and stabilizing shampoo using Adam. 2024.
69. Ribeiro MT, Singh S, Guestrin C. Why should I trust you?: explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. San Francisco, California, USA: ACM; 2016. pp. 1135–1144. https://doi.org/10.1145/2939672.2939778
70. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. In: Proceedings of the 31st international conference on neural information processing systems. NIPS'17. Red Hook, NY, USA: Curran Associates Inc.; 2017, pp. 4768–4777
71. Karniadakis GE, Kevrekidis IG, Lu L, Perdikaris P, Wang S, Yang L. Physics-informed machine learning. Nat Rev Phys. 2021;3(6):422–40. https://doi.org/10.1038/s42254-021-00314-5.
72. Vu MA, Yeung E, Li J-S. Data-driven control of oscillator networks with population-level measurement. Chaos Interdiscip J Nonlinear Sci. 2024;34(3):033107. https://doi.org/10.1063/5.0180436.

73. Bomela W, Singhal B, Li J-S. Engineering spatiotemporal patterns: information encoding, processing, and controllability in oscillator ensembles. Biomed Phys Eng Express. 2023;9(4):045033. https://doi.org/10.1088/2057-1976/acd9ea.

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